PREP 3, LESSONS 5–6

Springs, swings and conservation

Elastic potential energy; conservation of energy; energy dissipated

SCORE / 10

Name _____ Class _____ Due _____

Do Now REMEMBER

From earlier lessons, 1 mark each

1 State the standard unit of spring constant. DN 352 [1]

Newton per metre (N/m)

2 Name the store of energy associated with changes in height. DN 10 [1]

The gravitational potential energy store

3 Name the energy store that increases when an elastic object is stretched or compressed. DN 356 [1]

The elastic potential energy store

Key ideas LESSONS 5–6

Show your working

4 A spring with spring constant 40 N/m is stretched by 0.10 m. Use $E_e = \frac{1}{2} k e^2$

Calculate the energy in the elastic potential store of the spring. [3]

$E_e = \frac{1}{2} k e^2$ ✓

$E_e = \frac{1}{2} \times 40 \times 0.10^2$ ✓

$E_e = 0.20\,\text{J}$ ✓

5 A pendulum is pulled to one side and released.

a Describe the energy transfer as the pendulum swings down to its lowest point. [2]

Energy is transferred from the gravitational potential store ✓

to the kinetic store of the pendulum ✓

b It never quite reaches its starting height on the other side. Explain why, even though energy is conserved. [2]

Some energy is dissipated (by air resistance / friction at the pivot) ✓

to the thermal store of the surroundings; the total is still the same ✓